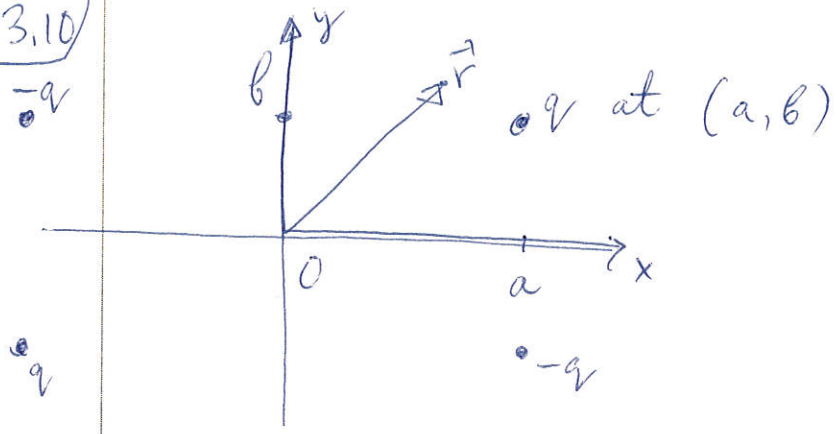
HW05

P3.10



Need 3 image charges: $-q$ at $(-a,b)$ and $(a,-b)$
and q at $(-a,-b)$

Electric potential at point $\vec{r}$ in the first quadrant:

$$V(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{|\vec{r}-(a\hat{x}+b\hat{y})|} - \frac{1}{|\vec{r}-(-a\hat{x}+b\hat{y})|} - \frac{1}{|\vec{r}-(a\hat{x}-b\hat{y})|} + \frac{1}{|\vec{r}-(-a\hat{x}-b\hat{y})|}\right\}$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{\sqrt{(x-a)^2+(y-b)^2}} + \frac{1}{\sqrt{(x+a)^2+(y+b)^2}} - \frac{1}{\sqrt{(x+a)^2+(y-b)^2}} - \frac{1}{\sqrt{(x-a)^2+(y+b)^2}}\right\}$$

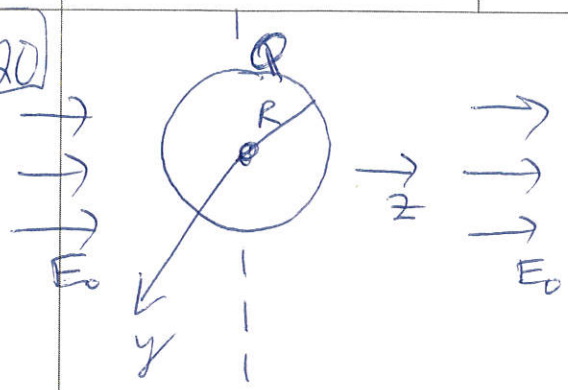
Check that lines $y=0$ and $x=0$ at zero potential (grounded)

$$V(x,y=0) = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\sqrt{(x-a)^2+b^2}} + \frac{1}{\sqrt{(x+a)^2+b^2}} - \frac{1}{\sqrt{(x+a)^2+b^2}} - \frac{1}{\sqrt{(x-a)^2+b^2}}\right) = 0$$

$V(x=0,y) = 0$ too. Force acting on q is then found from images as that

4 x

P3.20



as in Example 3.8, set $V=0$ in the xy plane (at $z=0$), for $\underline{x, y \to \infty}$

The sphere is not grounded, and has charge Q $\Rightarrow$

Boundary condition 1) $V = \text{const} = V_0$ for $r = R$ (equipotential)

2) $V = -E_0 r\cos\theta + \dfrac{Q}{4\pi\varepsilon_0 r}$

for $r \gg R$.

General form $V(r,\theta) = \sum_{l=0}^{\infty}\left(A_l r^l + \dfrac{B_l}{r^{l+1}}\right) P_l(\cos\theta)$ (eq. 3.65)

(1) $\Rightarrow \sum_{l=0}^{\infty}\left(A_l R^l + \dfrac{B_l}{R^{l+1}}\right) P_l(\cos\theta) = V_0$

$$\Rightarrow \begin{cases} A_0 + \dfrac{B_0}{R} = V_0 & (l=0) \quad B_0 = R(V_0 - A_0) \\ A_l R^l + \dfrac{B_l}{R^{l+1}} = 0 & (l \geq 1) \end{cases}$$

so that there is no angular dependence on the surface of the equipotential sphere.

$\Rightarrow B_l = -A_l R^{2l+1}$ for $l \geq 1$.

then $V(r,\theta) = A_0 + \dfrac{B_0}{r} + \sum_{l=1}^{\infty} A_l\left(r^l - \dfrac{R^{2l+1}}{r^{l+1}}\right) P_l(\cos\theta)$

(2) for $r \gg R$ $R^l\left(\dfrac{R}{r}\right)^{l+1} \to 0$ $\Rightarrow$ condition (2) becomes (all $l \geq 2$ are absent)

$$A_0 + \frac{B_0}{r} + A_1 r \underbrace{P_1(\cos\theta)}_{\cos\theta} = -E_0 r\cos\theta + \frac{Q}{4\pi\varepsilon_0 r}$$

$$\Rightarrow A_0 = 0 \;,\; B_0 = \frac{Q}{4\pi\varepsilon_0} \;,\; A_1 = -E_0 \;\left(\Rightarrow B_1 = E_0 R^3\right)$$

Thus the end result is

$$V(r,\theta) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r} + \left(-E_0 r + \frac{E_0 R^3}{r^2}\right)\cos\theta$$

$$= \frac{Q}{4\pi\varepsilon_0 r} - E_0\left(r - \frac{R^3}{r^2}\right)\cos\theta$$

which can be guessed from superposition principle:

$V =$ (potential of charged sphere) + (potential of the cond. sphere in external field E_0)